

$$n = 7$$

k num crossings	$k=1$	$k=2$	$k=3$
1	6	6	6
2	≥ 3	≥ 0	≥ 3
3	2	0	2

$$n = 8$$

k num crossings	$k=1$	$k=2$	$k=3$	$k=4$
1	7	7	7	7
2	≥ 4	≥ 2	≥ 2	≥ 4
3	≥ 3	≥ 0	≥ 0	≥ 3
4	2	0	0	2

$$n = 9$$

k num crossings	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
1	8	8	8	8	8
2	≥ 5	≥ 3	≥ 0	≥ 3	≥ 5
3	≥ 4	≥ 2	≥ 0	≥ 2	≥ 4
4	≥ 3	≥ 0	≥ 0	≥ 0	≥ 3
5	2	0	0	0	2

$\uparrow$
 3 chords
 2 chords
 1 chord
 0 chords

$$n = 10$$

k num crossings	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
1	9	9	9	9	9	9
2	≥ 6	≥ 4	≥ 2	≥ 2	≥ 4	≥ 6
3	≥ 5	≥ 3	≥ 0	≥ 0	≥ 3	≥ 5
4	≥ 4	≥ 0	≥ 0	≥ 0	≥ 0	≥ 4
5	≥ 3	≥ 0	≥ 0	≥ 0	≥ 0	≥ 3
6	2	0	0	0	0	2

$n = 11$

num crossings	k	k=1	k=2	k=3	k=4
1		10	10	10	10
2		≥ 7	≥ 5	≥ 3	≥ 0
3		≥ 6	≥ 4	≥ 2	≥ 0
4		≥ 5	≥ 2	≥ 0	≥ 0
5		≥ 4	≥ 0	≥ 0	≥ 0
6		≥ 3	≥ 0	≥ 0	≥ 0
7		2	0	0	0

Attempt: can we prove that an analogue for the n -gon with only l chords ($l \leq n-4$) vanishes if we impose $l+2$ distinct 1-zeros

$l=0, 1$ already proved

(NO)

assume we have two 1-zeros separated by a $+3$ shift (assume w.l.o.g. that they are $z_{1,0}$ and $z_{1,3}$)

start by ignoring $z_{1,0}$.

what survives? Things that have either 13 or 2n.

$$\rightarrow \frac{a_{13,2n}}{x_{13}x_{2n}} + \sum_{j=4}^{n-1} \left(\frac{a_{13,1j}}{x_{13}x_{1j}} + \frac{a_{2n,1j}}{x_{2n}x_{1j}} + \frac{a_{13,2j}}{x_{13}x_{2j}} + \frac{a_{2n,2j}}{x_{2n}x_{2j}} + \frac{a_{1j,2j}}{x_{1j}x_{2j}} \right)$$

surviving pairs for $z_{1,0}$

$(13, 2n)$
 $(13, 1j)$
 $(13, 2j)$
 $(2n, 1j)$
 $(2n, 2j)$
 $(1j, 2j)$

$j=4, \dots, n-1$

now look at $z_{1,3}$ now neither 13 or 2n are special (special are x_{46}, x_{35})

is $1j$ or $2j$ special? No

surviving pairs for $z_{1,3}$

$(46, 53)$
 $(46, 4j)$
 $(46, 5j)$
 $(53, 4j)$
 $(53, 5j)$
 $(4j, 5j)$

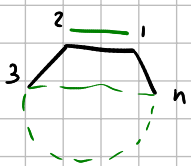
$j=7, \dots, n+2$

included (1,2)

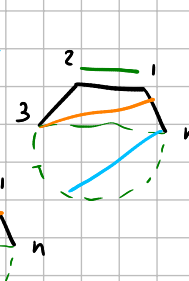
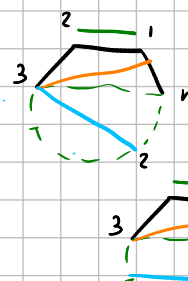
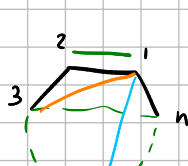
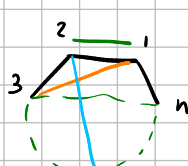
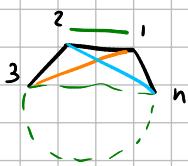
As long as $n \geq 6$ this works

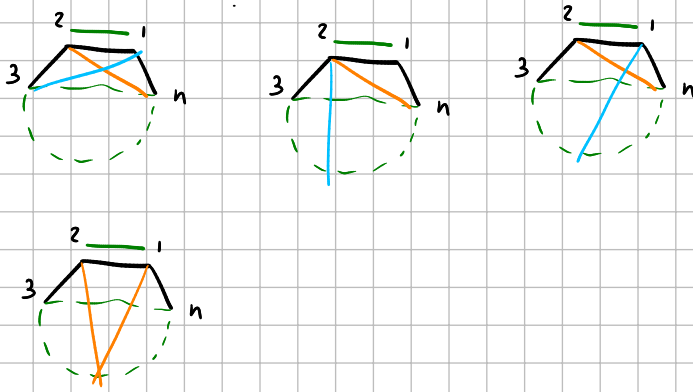
same idea w/ pictures

overlap: empty ✓



1-zero implies that the subgraphs are

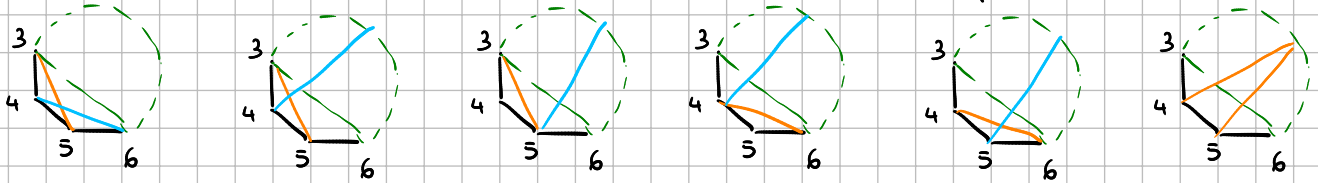




$(13, 2n)$
 $(13, 1n)$
 $(13, 2j)$
 $(2n, 1n)$
 $(2n, 2j)$
 $(1n, 2j)$

$j = 4, \dots, n-1$

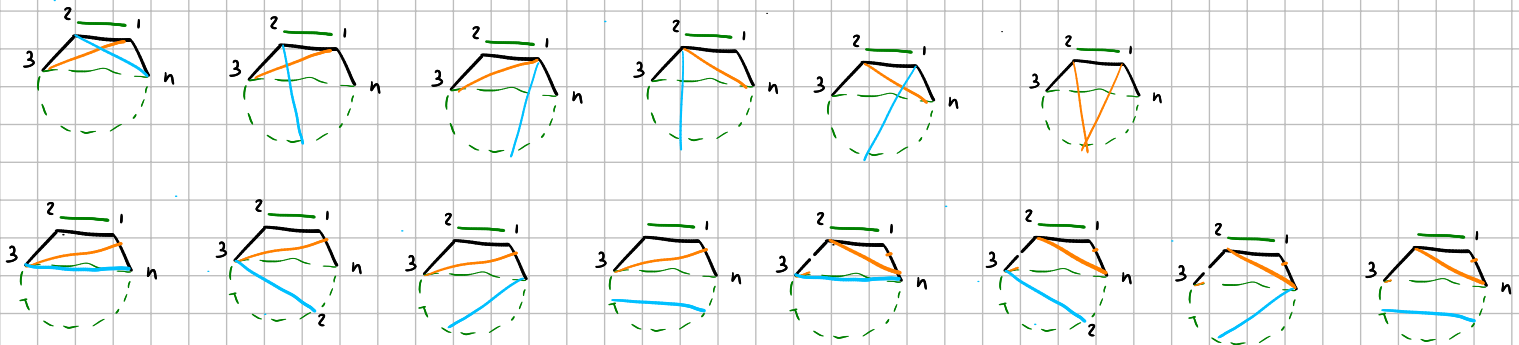
Diagram that survive $Z_{1,3}$



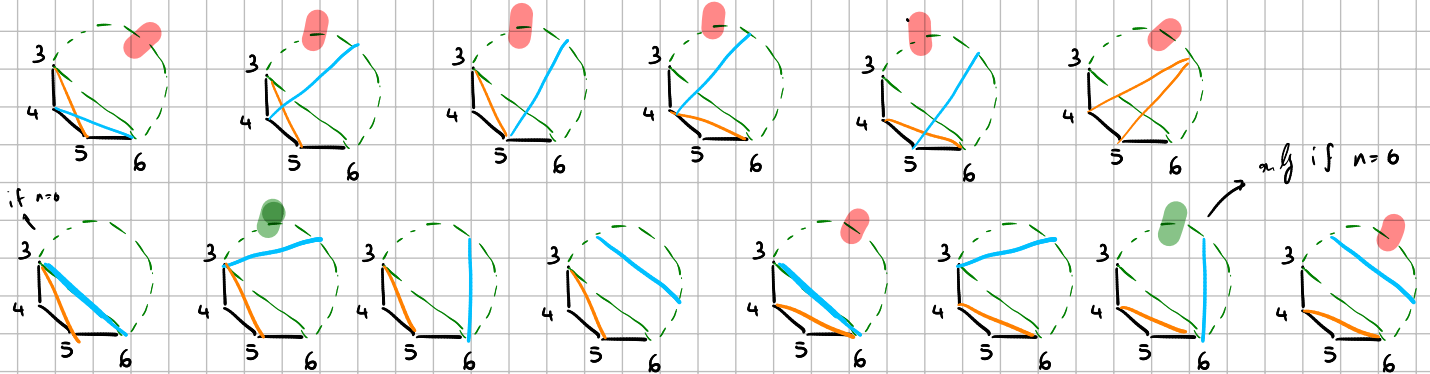
we are intersecting survivors of $Z_{1,0}$ and $Z_{1,3}$ and we see it's empty.

Note: this relies on the assumption of 2 chords in total and $n \geq 6$

full list $Z_{1,0}$



survive $Z_{1,3}$



intersection

Z_{1.0}

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{15} - X_{13}$$

$$X_{26} = -X_{13}$$

survivors

13	14	26	14	14	24
13	15	26	15	15	25
13	24	26	24		
13	25	26	25		
13	26				
13	35	26	35		
13	36	26	36		
13	46	26	46		

$$8 + 7 + 2 = 17$$

Z_{1.3}

46	41	53	41	41	51	x
46	42	53	42	42	52	x
46	51	53	51			
46	52	53	52			
46	53					
46	62	53	62			
46	63	53	63			
46	13	53	13			